

Part 1 (multiple choice): Fill in the above table with the correct answer:

Q 1) $\log \frac{15}{3} + \log \frac{3}{8} + \log \frac{8}{6} - \log \frac{5}{6} =$

a) 1

b) 2

c) $\log 3$

d) $\log \frac{10}{100}$

Q 2) The solution(s) for the equation $(\ln(ex))^2 + 2\ln(ex) + 1 = 0$ is (are)

a) $\emptyset$

b) e^{-3}

c) e^{-2}

d) e^{-3}, e^{-2}

Q3) $\frac{d}{dx} \left(\ln \left(\frac{x^2 e^{2x}}{\sqrt{1+3x}} \right) \right)$, at $x = 1$ equals:

a) $\frac{13}{4}$

b) $\frac{11}{4}$

c) 12

d) $\frac{16}{4}$

Q4) $\frac{d}{dx} ((\ln x)^{\ln x})$, at $x = e$

a) e

b) 0

c) $\frac{1}{e}$

d) $\frac{2}{e}$

Q5) $\lim_{x \rightarrow \infty} \left(1 + \frac{2}{x} \right)^{-6x} =$

a) e^{12}

b) e^{-12}

c) e^{-6}

d) does not exist

Q6) $\sec(\tan^{-1} 2x) =$

a) $\sqrt{1+x^2}$

b) $\sqrt{1+2x^2}$

c) $\sqrt{1+3x^2}$

d) $\sqrt{1+4x^2}$

Q7) $\int \frac{1}{25+x^2} dx$

a) $\frac{1}{5} \tan^{-1} \frac{x}{5} + c$

b) $\tan^{-1} \frac{x}{5} + c$

c) $\frac{1}{5} \tan^{-1} x + c$

d) $-\frac{1}{5} \tan^{-1} \frac{x}{5} + c$

Part 2: Written question.
Evaluate the following integrals

1) $\int x^4 e^{-2x} dx$ $x^4 \rightarrow 4x^3$
 $\int dv \rightarrow \frac{e^{-2x}}{-2}$ (3)

1) $\int x^4 e^{-2x} = \frac{e^{-2x}}{-2} \times x^4 + \int 4x^3 \frac{e^{-2x}}{-2} dx$

تکامل
اجزاء

2) $\int \frac{4x^3}{2} e^{-2x} \rightarrow \frac{2x^3}{2} e^{-2x} + \frac{1}{2} \int 2x^2 e^{-2x}$

$\int x^4 e^{-2x} = \int \frac{x^4}{-2} e^{-2x} - \frac{2x^3}{2} e^{-2x} + \int \frac{x^2}{2} e^{-2x}$ تکامل
اجزاء

$\frac{x e^{-2x}}{-2} - \frac{2x^3 e^{-2x}}{2} + \frac{x^2 e^{-2x}}{-2} + \frac{x e^{-2x}}{-2} + \int e^{-2x}$

$\frac{x e^{-2x}}{-2} - \frac{2x^3 e^{-2x}}{2} + \frac{x^2 e^{-2x}}{-2} + \frac{x e^{-2x}}{-2} - \frac{e^{-2x}}{2} + C$

2) $\int_0^{2\pi} \cos(mx) \cos(nx) dx$ (3)

$(\cos(m-n)x + \cos(m+n)x)$

$\int_0^{2\pi} \frac{\cos(m-n)x}{2} + \int_0^{2\pi} \frac{\cos(m+n)x}{2}$

$\int_0^{2\pi} \frac{\sin(m-n)x}{2(m-n)} + \int_0^{2\pi} \frac{\sin(m+n)x}{2(m+n)}$

$$\left(\frac{\sin(m-n)2\pi}{2(m-n)} - \frac{\sin(m-n)0}{2(m-n)} \right) + \left(\frac{\sin(m+n)2\pi}{2(m+n)} - \frac{\sin(m+n)0}{2(m+n)} \right) = 0$$

3) Evaluate $\lim_{x \rightarrow 0^+} \ln(\ln(\sin 2ex)) - \ln(\ln(\tan x))$

$$\lim_{x \rightarrow 0^+} \ln \left(\frac{\ln \sin 2ex}{\ln \tan x} \right) = \frac{0}{0} \quad (2)$$

$$\frac{\ln \sin(e^x)^2}{\sec^2 x} \quad \lim_{x \rightarrow 0} \ln \frac{2 \ln \sin e^x}{\sin 2e^x} \cdot \frac{\sec^2 x}{\tan x}$$

$$\frac{\ln \sin e^x}{\ln \tan x} \quad \frac{2 \cos 2e^x (2e^x)}{\sin 2e^x} \cdot \frac{\sec^2 x}{\tan x}$$

Q34

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$$\frac{\cos 2e^x (2e + 2e^x)}{}$$

Good Luck